Data Recovery in Linux

Part—1

In the first part of this series, I will cover how to perform data recovery in Linux, i.e., getting your data back in case you formatted the drive, or lost data due to any other reason.

About a month back, a colleague, Sandeep C V, mistakenly formatted his external hard disk, which contained precious data. So I spent the next few hours searching for a solution to his problem. I stumbled on a lot of proprietary software that claimed to fix things but was bent on finding an open source application, being a FOSS advocate. After hours of search, I finally succeeded! A company called CGSecurity had two utilities, TestDisk and PhotoRec, which are free to use. You can get them at *http://www.cgsecurity.org/wiki/TestDisk_Download*.

On the right is a list of dos and don'ts that must be followed in case you have lost data (on external media, on the internal hard disk of your laptop, etc).

The don'ts	The dos
Don't write any more data to the device. Shut down the system immediately.	Reboot the system with a live CD or other live version of Linux running off other media, like a USB thumb drive.
Don't attempt to perform a recovery on your own if you are a novice.	Follow the instructions in this article.
Don't mount the media that has the lost data just yet.	Download TestDisk or PhotoRec if not in the live CD (see tip below).
Don't use any random software that claims to recover data.	Pray that all goes well!

File deletion—what happens?

Usually, when we delete a file, it goes to the Trash/Recycle Bin, provided the file is not larger than the bin's capacity. One can restore these files easily. If you have emptied the Recycle Bin, permanently deleted the file using *Shift+Delete* or an *rm* command at the command-line, you may think the file is irretrievable—it isn't. Only a pointer to the file is deleted as yet, not the data, which remains on the disk.

Data stored on a medium as files has a 'table of contents' indicating the storage location for each file (name) on the drive. When a file is 'deleted', its entry is removed from the table of contents. This indicates that the space previously used by the file can be used to store other data—it is now 'available' space. When a new file is written to the disk, and uses this 'available' space, it replaces the previous data with new data. Now, recovery of the old file becomes very difficult (though